

Theme 1 - Functions

Note Title

2014/01/24

Unit 1.1 Numbers, intervals & inequalities pA2

Number & intervals - Self study.

Inequalities page A4:

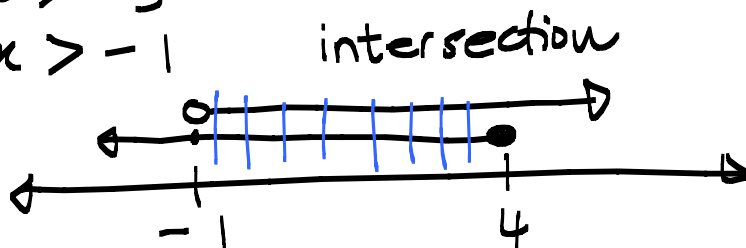
1. Solve $3x + 4 \leq 16$ ←
 $3x \leq 12$
 $x \leq 4$

$\{x \mid x \leq 4\}$
Set builder notation

$x \in (-\infty, 4]$
Interval notation

2. $1 < 3x + 4 \leq 16$

$\Rightarrow 1 < 3x + 4$ and $3x + 4 \leq 16$
 $3x > -3$ $\cap$ $\therefore x \leq 4$
 $x > -1$



$\therefore x \in (-1, 4]$

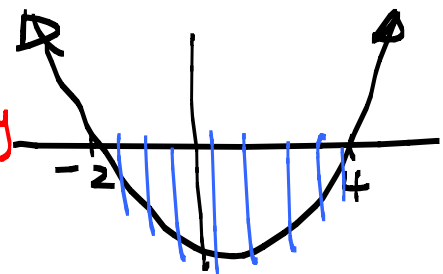
3. $x^2 < 2x + 8$
 $x^2 - 2x - 8 < 0$
 $(x+2)(x-4) < 0$
 ↑ ↑

		-2		4		
$x+2$		-		+		+
$x-4$		-		-		+
x		+		-		+

$$x \in (-2, 4)$$

$$x \in (-2, 4)$$

Inequality



Intervals are a pair.

$$4. \downarrow x^3 + 3x < 4x^2$$

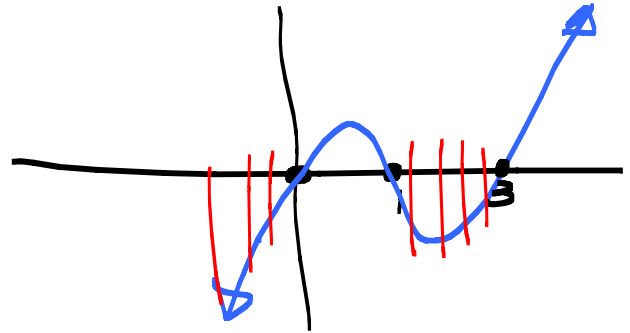
$$x^3 - 4x^2 + 3x < 0$$

$$x(x^2 - 4x + 3) < 0$$

$$x(x - 1)(x - 3) < 0$$

$$x \in (-\infty, 0) \cup (1, 3)$$

union
or



$$5. \frac{1}{x} < 4$$

!!!

$$x^2 \times \frac{1}{x} < 4 \times x^2$$

$$x < 4x^2 \quad (x \neq 0)$$

$$4x^2 - x > 0$$

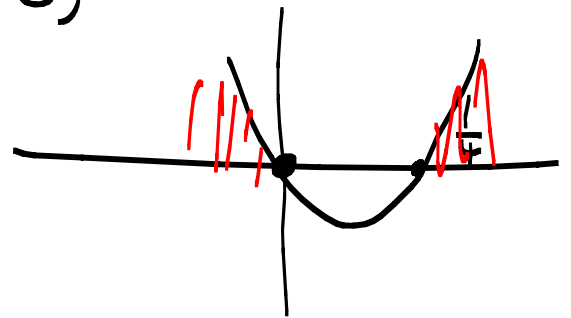
$$x(4x - 1) > 0$$

$$x \in (-\infty, 0) \cup (\frac{1}{4}, \infty)$$

$$\frac{1}{x} - 4 < 0$$

$$+ - \frac{1 - 4x}{x} < 0$$

$$- +$$



$$6. -3 < \frac{1}{x} \leq 1 \quad (x \neq 0)$$

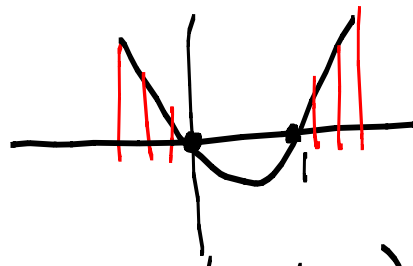
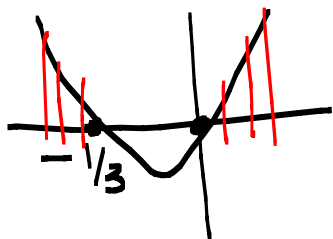
$$\Rightarrow -3 < \frac{1}{x} \quad \text{and} \quad \frac{1}{x} \leq 1$$

$$\Rightarrow -3x^2 < x \quad \text{and} \quad x \leq x^2 \quad (x \neq 0)$$

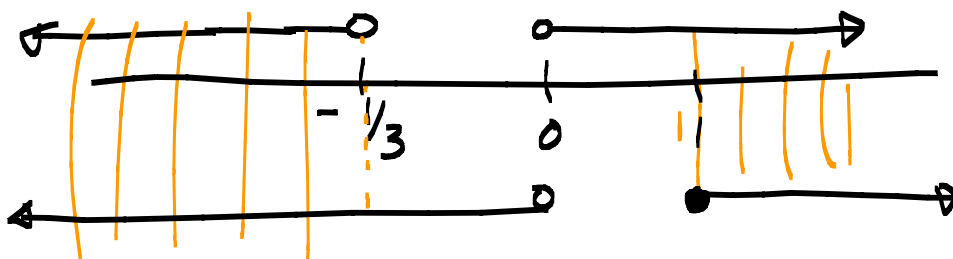
$$\Rightarrow -3x^2 - x < 0 \quad \text{and} \quad x^2 - x \geq 0$$

$$\Rightarrow 3x^2 + x > 0 \quad \text{and} \quad x^2 - x \geq 0$$

$$\Rightarrow x(3x + 1) > 0 \quad \text{and} \quad x(x - 1) \geq 0$$



$$\Rightarrow x \in (-\infty, -\frac{1}{3}) \cup (0, \infty) \quad \underline{\underline{\text{and}}} \quad x \in (-\infty, 0) \cup [1, \infty)$$



$$x \in (-\infty, -\frac{1}{3}) \cup [1, \infty)$$

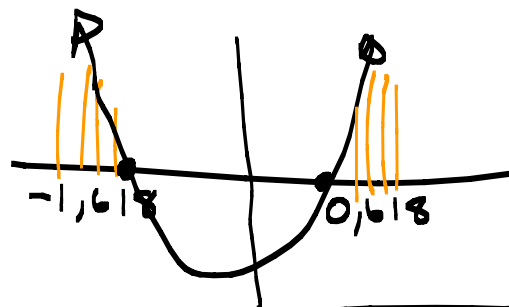
7. $\frac{x+1}{2-x} \geq 0$ Answer: $x \in [-1, 2)$

8. $x^2 + x - 1 > 0$! Inequality.

$(\quad) > 0 ??$

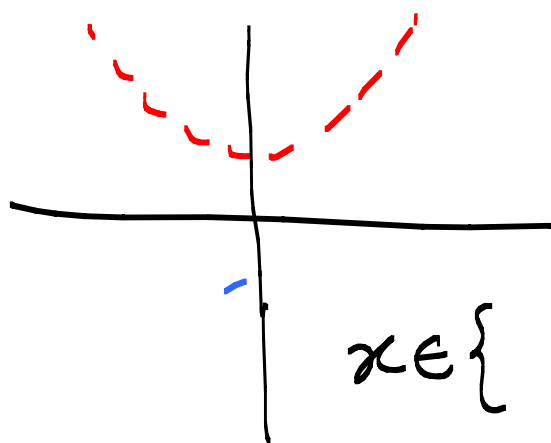
$$x^2 + x - 1 = 0 \Rightarrow x = -\frac{1}{2} \pm \frac{\sqrt{5}}{2}$$

$$x \in (-\infty, -\frac{1}{2} - \frac{\sqrt{5}}{2}) \cup (-\frac{1}{2} + \frac{\sqrt{5}}{2}, \infty)$$



9. $-x^2 - x - 1 > 0$ ✗

$$\Rightarrow * x^2 + x + 1 < 0 \quad \leftarrow$$



$$x = \frac{-1 \pm \sqrt{1 - 4(1)(1)}}{2}$$

$$\Delta < 0 \quad [b^2 - 4ac]$$

$\Rightarrow$ no real roots.

$$x \in \{ \} \quad \text{or} \quad x \in \emptyset$$